Supplement to: Capability or Optimizer? What Lets an LLM Proposer Leave Its Template Family on Circle Packing

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Section, table and figure references without an S prefix point into the main paper. Each arm here has a row in the paper's Table 1 and a ledger listed in Appendix D; the paper's deviations table (Table 3) covers these arms too. Nothing here is pooled with any row of Table 2.

S1 The pooled zero over six arms

Table 2's zeros pool with three arms outside it that test the same value in regimes the direct rows do not cover — held-out cells, one conditioning step, five loop generations — and the pooled statement is the one an evaluation reader can use:

corpus	regime	clears	where
115 valid program outputs (CC 37, CC2 41, CCS 37)	executed math-only code, two tiers, fresh-draw replication	0	§3.4
63 valid outputs	held-out N, five cells absent from every scoreboard	0	Appendix E.5
49 valid conditioned outputs	five generations of a registered archive-and-mutate loop	0	§5.2
63 valid conditioned outputs	one conditioning step, three parent conditions (10^{-9})	0	§5.1
290 valid outputs (281 at 10^{-9} throughout)	CC, CC2, CCS, CN, MU, L	**0**	—
12 valid program outputs	math-only code, bare pinned API path (not pooled)	0	§5.3
35 valid program outputs	math-only code, Sonnet tier, pinned path, 120 s (not pooled)	0	§3.5
19 valid program outputs	numpy/scipy code, weak tier, arms CL and CL-W pooled (not pooled here)	0	§3.5
25 valid program outputs	numpy/scipy code, Sonnet tier (not pooled)	**23**	§3.5

Pooled, that is **0 of 290 valid outputs** across six arms, selected by regime and not by outcome: math-only programs at two tiers, cells no scoreboard lists, a conditioning step, an iterated loop. Three things travel with the number. It is not one tolerance: the first three rows count outputs valid at 10^{-6}, the arm MU row counts at 10^{-9}, and arm MU is the one arm where the legs disagree — held at 10^{-6} throughout, the pool reads 315 valid outputs with 18 above the argmax, all 18 from arm MU and every one failing 10^{-9} because the excess is its own overlap (§5.1). Held at 10^{-9} throughout, which is the tolerance the clearance rule requires, the pool reads 281 valid outputs with 0 above: the three code arms are unchanged (37, 41 and 37 valid at both tolerances, re-scored from the stored programs through the registered scorer), CN 56 of its 63, L 47 of its 49, MU 63. The recount is post hoc bookkeeping over rows already in the study (`recount_1e9.py`); it adds no invocations and moves no verdict. It is not all registered: arms CC/CC2/CCS registered zero-count predictions and arm L's L4 was registered open, but the clearance question was put to arm MU's rows post hoc; arm CH's 14 valid outputs, conditioned on a score table rather than on a parent, are held out of the parent-conditioning row and none of them clears either. And it is not a bound: the three direct-emission escapes of §3.3 and the twenty-three library programs of §3.5 sit outside it by named regime, and the middle-tier escape sits at $N = 31$, a cell five of the six pooled arms also sample. What the zero says is that on this benchmark, at these cells, a weak-tier proposal or a math-only program at either tier does not exceed a value computable in advance, so a discovery claim from those proposers has a number to beat before any search is credited.

S2 The cross-vendor arms in full

§5.4 states the verdicts; this section carries the chain's verdict table, arm GM3 per cell, and arm V in full.

Vendor	Model	Instrument	Registered test	Verdict
Google	gemma-4-26b	first-party API, 7 N × 20 (41/140 rows routed under amendment; dual accounting)	pooled ≥ 30%; cell-level ≥ 5; falsifier ≥ 4 fails	pooled bar met (57.5%); discriminating cells 11/43 = 26%, **below** the bar; cell-level short (modal in 2 of the 5 scoreable cells, against a bar of 5; falsifier not triggered); all three misses are the *rival construction* (27/43)
Cohere	north-mini-code	free-tier alias, 5 N × 5	V1 ≥ 50% of valid on-prediction; V2 rival ≤ 10%	**TRANSFERS**, point-estimate (8/12 = 67%; CI includes bar); V2 HOLDS (0/4)
OpenAI (open-weight)	gpt-oss-20b	free-tier alias, 5 N × 5	same	**TRANSFERS**, point-estimate (11/18 = 61%; CI includes bar); V2 HOLDS (0/9)
Google	gemma-4-31b	free-tier alias, 5 N × 5	same	**DOES-NOT-TRANSFER** (0/15); V2 HOLDS (0/3)
10 further aliases	(NVIDIA, Liquid, Poolside, inclusionAI, others; one unattributed)	free-tier	floor: ≥ 10 of 25 invocations valid	BELOW-FLOOR; failure taxonomies logged, no claim in either direction

S2.1 Arm GM3 per cell

Arm GM3. Arm GM3 asks whether the anchoring law is a fact about the original lineage or about weak-tier models as a class, by re-running the unconditioned-call protocol against a second vendor's open-weight family (gemma-4-26b-a4b-it) through its first-party API: seven N cells, twenty samples each, registered predictions and falsifier fixed before sampling (prereg chain in Appendix D; the instrument's history — a first attempt whose 4,096-token budget was consumed entirely by the model's deliberation channel, resolved by a registered re-run at 16,384 — is in §7, and is itself a finding about output-budget confounds in cross-vendor comparisons). Per cell, recomputable from `arm_gm_gm3_checkpoint.jsonl` by `arm_gm3_analysis.py` with no arguments:

N	predicted (4 dp)	Gemma modal sum	mode freq	valid	on-prediction	MODE-MATCH
13	1.6250	1.7761	7/9	9	0	no (misses upward)
17	2.0518	**2.0518**	19/20	20	20	**yes**
21	2.1000	2.2589	9/18	18	8	no (misses upward)
31	2.5833	2.7485	11/16	16	3	no (misses upward)
35	2.9167	**2.9167**	15/15	15	15	**yes**
37	3.0345	—	—	2	0	UNSCOREABLE
43	3.0714	—	—	0	0	UNSCOREABLE

Three registered outcomes (Figure S1). **The pooled prediction held**: 46 of 80 valid packings (57.5%) land within 2×10^{-3} of the value predicted by §2's rule, against a registered 30% bar — for a rule fitted entirely on another vendor's models, before this family was ever sampled. The registered metric pools all cells, and the pass rides on that: by Appendix E.2's own convention — arm F's pooled rate excludes non-discriminating cells *because prediction equals the family argmax there* — the decomposition must be stated. 35 of the 46 on-prediction packings sit in the two non-discriminating cells (20/20 at $N = 17$, 15/15 at $N = 35$); on the three scoreable discriminating cells the rate is 11 of 43 (25.6%), below the bar the pooled metric passes. The registered verdict stands as registered; what it measures at the discriminating cells is the next finding. **The cell-level prediction did not hold**: the predicted value is modal in 2 of 5 scoreable cells, short of the registered 5, though the registered falsifier (failure in ≥4) also did not trigger — and where the family locks

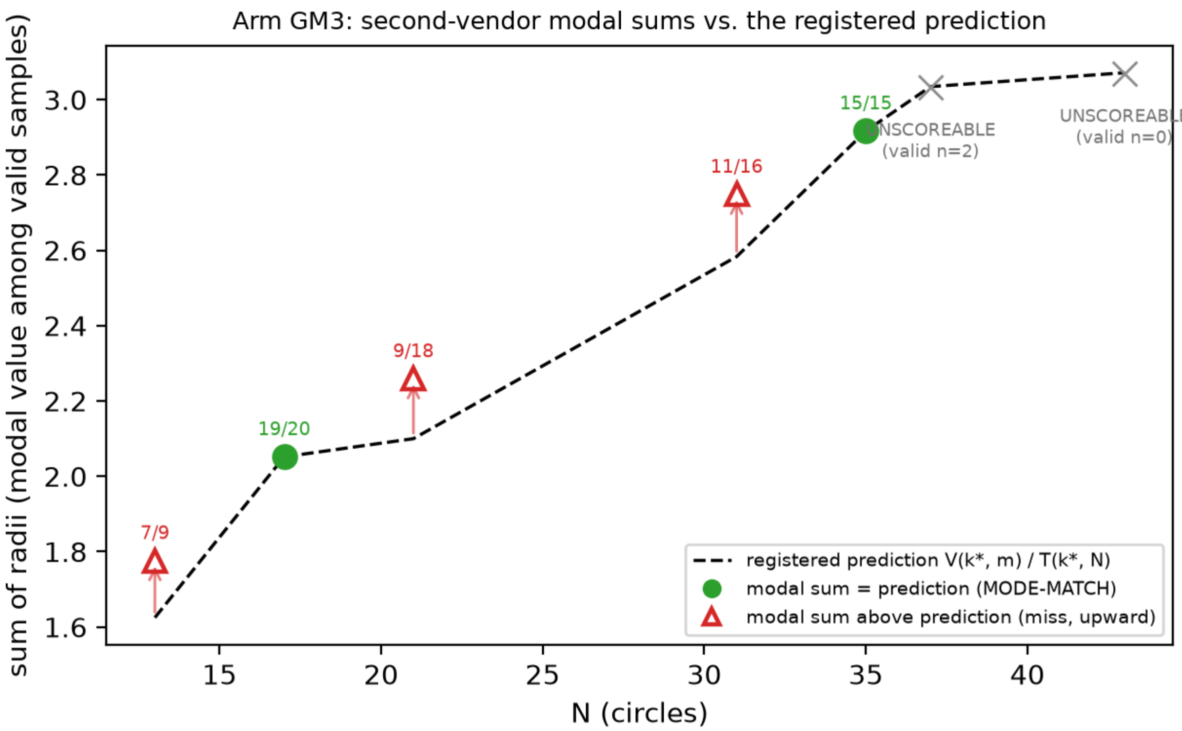

Figure S1: *Arm GM3 modal sums against the registered prediction.* Filled green circles match the predicted mode (the dashed line is $V(k^*, m)$ at extend cells and $T(k^*, N)$ at truncate cells) (frequencies annotated); open red triangles miss — every miss lands above the prediction, on the rival argmax (the modal packings carry the rival's two-radius structure, 27/27 — §5.4); the two largest cells are UNSCOREABLE under the registered <3-valid rule. Regenerates via `fig_gm3_anchoring.py` from `arm_gm3_report.json` (itself replayed from the raw ledger by `arm_gm3_analysis.py`, no arguments).

on, it locks on almost completely. That pair of bars leaves a dead zone: cell-level outcomes of 2, 3 or 4 confirm nothing and falsify nothing, and the observed outcome landed in it. The registration is at fault, not the result; we report the verdict it defines and note that a registration with a gap this wide buys less than it appears to. **The structural signature came in under its bar**: 20 of 46 on-prediction samples (43%, bar: half) carry the exact two-radius grid-plus- filler decomposition. The instrument is blind by construction at truncate cells, whose prediction is single-radius: all 20 signatures come from $N = 17$ (20 of 20), the one extend cell among the matches, while $N = 35$'s 15 on-prediction truncations carry no filler to detect (0 of 15). The registered verdict stands as registered; read per eligible cell, the construction matches wherever it can be seen.

The three cell-level misses are not noise, and "misses upward" understates them: **the family's modal output at every discriminating cell is the rival argmax itself** — 1.7761, 2.2589 and 2.7485 against rivals 1.7761424, 2.2588835 and 2.7485281 — and a post-hoc structural diagnostic (disclosed as such; `diagnostics_gm3_rival.py`, read-only over the ledger) finds all 27 of 27 rival-valued packings carrying the rival's exact two-radius decomposition, the (k^*-1) grid radius with the (k^*-1) filler radius. The registered signature instrument could not see this: it tests radii against the *predicted* order k^*, so it is structurally blind to (k^*-1) constructions — which is why the 43% signature figure above coexists with a 100% rival-structure rate on the miss side. In §2.3's terms, gemma-4-26b performs precisely the drop-to-(k^*-1)-and-fill move that "a proposer with any local search would" perform and that the original family performs in 2 of 57 valid samples: 27 of 43 valid discriminating-cell packings (63%) are the rival construction. Read against §7's tractability alternative, the anchor value marks where construction-by-mental-arithmetic lands, and a family with more arithmetic reach executes the harder in-family construction rather than truncating — same recipe family, other branch. Validity collapsing at the two largest cells (2 then 0 valid of 20) bounds the family's competence window on this task; both cells are UNSCOREABLE under the registered rule (< 3 valid samples, fixed in the GM-chain registration before sampling) and claim nothing.

S2.2 Arm V in full: cross-vendor zero-shot under free-tier serving

§5.4 is the contract; this is the report.

Arm V. Arm GM3 varied the vendor while holding the instrument close to laboratory grade — first-party API, pinned temperature, explicit output budget, seven cells at twenty samples. Arm V varies vendor and serving class at once, under the same byte-identical arm-F prompts both arms inherit (SHA-256 per N restated in the registration and recomputed at call time, aborting on mismatch; the GM-chain registration pins the identical hashes), at five cells and five samples per (model, N), against free-tier aliases across seven attributed vendors (`arm_v_preregistration.md`, committed before sampling; three amendments, each committed before the sampling it governs, expanded the original three-alias design to thirteen aliases across two serving paths — the opencode CLI and OpenRouter chat-completions — and raised `max_tokens` to 24,576 after hidden-reasoning truncation emptied three models' replies). Serving disclosures per the companion paper's protocol: aliases on managed paths, `sampling_params: null` on every row, per-row upstream-provider strings logged where the path reports them, every invocation appended live to `arm_v_candidates_raw.jsonl` — 877 invocations, failures and refusals included. Scoring is `arm_f_repro.py`'s parse/validate/classify unchanged, tolerances registered in arm F before its sampling.

Registered floor: $\geq$10 of a model's 25 registered invocations valid at 10^{-6}; below it, no anchoring claim in either direction. Execution retried each invocation slot until content arrived, per amendment 3's registered transport-repair and accounting rule, so the ledger holds many more attempt rows than slots — gemma-4-31b's 25 slots, for instance, took 190 attempt rows, 165 of them transport failures — and the floor is evaluated on the 25 slots, not the attempts. Ten of thirteen aliases finished BELOW-FLOOR — including all three originally-registered opencode aliases, whose logged failure taxonomies are transport-dominated for one (CLI build banners, timeouts, "Unknown model" errors as the free tier rotated its catalogue) and roughly half transport, half parse-failure for the other two — plus one alias returning all 108 of its attempts without content. Three aliases cleared the floor, and their registered verdicts split:

alias (vendor)	valid	V1 on-prediction	V1 verdict (bar: $\geq$50%)	V2 rival-argmax (bar: $\leq$10%)
`north-mini-code` (Cohere)	12	8/12 [39–86%]	**TRANSFERS** (point est.)	0/4 → **HOLDS**
`gpt-oss-20b` (OpenAI, open-weight)	18	11/18 [39–80%]	**TRANSFERS** (point est.)	0/9 → **HOLDS**
`gemma-4-31b-it` (Google)	15	0/15 = 0%	**DOES-NOT-TRANSFER**	0/3 → **HOLDS**

This is the registration's most informative branch pair. Anchoring under this instrument is not an artifact of the original family: two further vendors' models land the registered construction values at 61–67% of valid outputs, point estimates above the 50% bar.

Both passes are weaker than those percentages suggest, in three ways we state rather than leave to the reader. The bracketed Wilson 95% intervals include the bar, so each verdict is a point-estimate pass at $n = 12$ and $n = 18$. Three of thirteen aliases were evaluable at all; the other ten finished BELOW-FLOOR and claim nothing in either direction. And the pooled rate spans cells that cannot discriminate anchoring from family search. Decomposing onto the discriminating cells alone — the split this section demands of the GM chain, and symmetry demands here — gives 3 of 4 and 5 of 9. Unlike GM3 the decomposition does not reverse either verdict, but it shows how little of each pass sits where prediction and family argmax differ.

Appendix E.2's template-shape null belongs here too, and applying it costs the arm its strongest reading. Against that null a proposer lands on-prediction one time in three. The pooled rates clear it — exact upper tails 0.019 and 0.014 — but they clear it on cells where prediction equals the family argmax. At the cells that can discriminate, neither pass separates: 3/4 gives 0.111 and 5/9 gives 0.145, and GM3's 11/43 gives 0.895, below the null. So the honest cross-vendor statement is that two further vendors are *consistent with* the behaviour and that at $n = 4$ and $n = 9$ the discriminating evidence does not separate them from a shape-uniform proposer. Recomputed in `cross_vendor_null.py`; post hoc, like the null itself.

The strong form holds everywhere evaluable *within this arm*: zero rival-argmax emissions in 16 discriminating-cell validities. The registration (`arm_v_preregistration.md`) recorded the original family's prior for the same statistic as 0/21 over $N \in \{13, 31\}$ only, a narrower scope than Appendix E.3's 2/23 whose two rival hits both sit at $N = 21$, so the two corpora agree at 0 on the registration's own cells. The standing exception is gemma-4-26b under the GM-chain instrument, whose discriminating-cell mode *is* the rival.

The transfer's point estimates sit at or below the in-family rates: 61–67% on-prediction against 56–76% in-family at the same five cells (Appendix E.2). The ranges overlap, so reduced concentration is a directional reading these n cannot separate. Directionally lower concentration at different vendors, on byte-identical prompts, landing on the same modal value is what a shared attractor predicts; equal concentration would be harder to separate from shared training text. And it is not universal: gemma-4-31b, clearing the same floor, lands the prediction never. Its valid packings are unanchored and low (bests 0.88–1.85 against predictions 1.63–3.03) — not the rival, not the template, just weaker de-novo construction. Under the registration's falsification map, the family-specific branch fires — any model clearing the floor with DOES-NOT-TRANSFER makes the closed form a family-bounded property, stated in §3.3 with data rather than a caveat — while the map's all-transfer branch fails on gemma alone, so the reading that anchoring is an artifact of the original family is defeated by the same table that bounds its generality.

The gemma pair needs stating precisely, because this family now appears on both sides, and the prompts are identical across the two arms — what differs is parameter count, serving path and sample budget, not the elicitation. Under the GM-chain instrument (first-party, pinned temperature, 7×20), gemma-4-**26b** met the pooled bar at 57.5% while its discriminating-cell rate (26%) fell below it (the mixed-serving accounting; first-party-only, 59.0% — §7's registered deviation applies to this figure) while emitting the *rival* construction at 63% of discriminating-cell validities; under arm V's free-tier instrument (5×5), gemma-4-**31b** through a rate-starved alias (Google AI Studio upstream; 27–38 of each cell's attempts returned as shared-pool 429s before content arrived) lands neither the prediction nor the rival — its valid packings are weaker de-novo constructions. The pair bounds the family's behaviour rather than contradicting itself — one member sits *above* the template's execution reach (far enough to build the drop-and-fill rival the original family cannot), the other lands below it — but model variant and instrument quality change together across the two measurements, so which difference carries the split is not attributable from this pair; either way, no single-model measurement would have seen the boundary.

S3 The `opus_alias` arm: the dual accounting

Appendix C carries the ladder table and the failure taxonomy; the two post-hoc decisions and the two corpora behind the ladder's first rung are here.

Two decisions here were made after seeing the data and are recorded as deviations (Table 3). P-O1, P-O2 and P-O4 are reported **not evaluable**, on the grounds that a validity collapse of this size makes an on-prediction tier comparison uninterpretable; P-O3 is trivially satisfied on 4/4 valid samples; the registered disconfirmation — regression toward the trap — did not occur. De-registering three of four predictions in the one arm that disconfirmed is exactly the move preregistration exists to constrain, which is why it is tabled rather than narrated. And the arm was initially excluded from the ladder, then included once its results were known; we follow the later decision and report the ladder both ways (Table 4) — with the arm 83% → 100% → 13% on matched cells, without it 83% → 100% plus the qualitative shift from truncation to perturbed multi-radius hybrids, which is already the publishable contrast.

Two rungs, two corpora, and they are easy to confuse. The 83% is the 60-row matched-cell ledger at $N \in \{13, 21, 31\}$. The 78% that circulates beside it is the bare arm's own 45-row ledger over all seven weak-tier cells — a different corpus and a different cell set, not a fourth rung. The 10^{-9} ladder reads 64% → 90% → 13%, where the 64% is again the 45-row seven-cell ledger and its matched-cell counterpart was not computed, so the two ladders are not directly comparable at their first rung either. Both rows appear separately in Table 4 and neither figure is quoted without its corpus.

S4 Arm M in full: the filler branch fails at $m > 1$

Appendix E.4 carries the registered verdicts and Table M; this is the full report. Two gaps remained after the square arm: (i) $V(k,m)$'s filler branch was empirically supported only at $m \in \{0,1\}$ — the converge cells $N = 17$ and 37 both have $m = 1$ — and (ii) the $k = 8$ trap zone appeared in Figure 2 but was never sampled. Arm M closes both gaps and was registered before sampling: $n = 15$ bare weak-tier invocations at each of $N = 20$ ($k^* = 4$, $m = 4$, predicted $V(4,4) = 2.2071068$), $N = 30$ ($k^* = 5$, $m = 5$, $V(5,5) = 2.7071068$), $N = 41$ ($k^* = 6$, $m = 5$, $V(6,5) = 3.1725890$) and $N = 57$ ($k^* = 8$, trap, $T(8,57) = 3.5625000$, rival $V(7,8) = 3.7366935$), with an explicit tie-stated falsifier for each half.

The falsifier for the filler branch triggered at 3 of 3 cells. The registered predictions P-M1–P-M3 are disconfirmed: not one of the three converge cells has $V(k^*,m)$ as its modal output. The model does not extend a k^*-grid with fillers; it moves *up* to the $(k^*{+}1)$ grid and truncates or exactly fills it — modal outputs $2.0 = T(5,20)$ (12 of 15 valid; $N = 20$ is the 5×4 rectangle exactly), $2.5 = T(6,30)$ (11 of 14; 6×5 exactly) and $2.9285714 = T(7,41)$ (6 of 7; 7×6 minus one). Exactly one sample in 36 valid converge-cell rows emitted the registered construction — a textbook $V(4,4)$, all four fillers at $(\sqrt{2}-1)/8$. Per the registered falsifier wording, $V(k,m)$ **is restricted to the** $m \leq 1$ **support previously observed**, and the branch rule's extend arm is disconfirmed at $m \geq 4$; the intermediate "places some fillers" outcome named in the registration did not occur (filler count 0 in 35 of 36 valid converge rows). The emergent regularity is arithmetic, not geometric: where N factors as $a \times b$ with a, b near $\sqrt{N}$ the model emits that exact rectangle of uniform circles ($20 = 5 \times 4$, $30 = 6 \times 5$), which is further evidence for §7's arithmetic-tractability alternative — rectangle factorization is mentally computable where filler tangency is not.

P-M4 confirmed. At $N = 57$ the modal valid output is $T(8,57) = 3.5625000$ (6 of 10 valid at 10^{-6}), the $k = 8$ grid truncated to 57; 0 of 10 valid samples reached the rival $V(7,8)$, against a registered rival prediction of 0. The branch rule's truncate arm now has five confirmed k (4, 5, 6, 7, 8) and Figure 2's $k = 8$ zone is sampled. Validity by cell: 15/15, 14/15 (one parse failure — fraction literals), 7/12 and 10/15 at 10^{-6}. Three $N = 41$ invocations died in the runtime before reaching a model and are excluded under the rejection rule registered for this arm in registration — registered precisely because the analogous arm F exclusion was not (§8) — and counted here: the cell is $n = 12$ sampled of 15 launched. Raw rows verbatim in `arm_m_collect.jsonl`; scoring in `arm_m_analysis.py`, conventions identical to arm F.

Table M is in Appendix E.4.

Exactly one $V(k^*,m)$ construction appears in 36 valid converge-cell rows; filler count is 0 in 35 of 36. Three converge-cell rows exceed the recipe family's best *upward*, of which two clear it under §2.4's rule — two at $N = 20$ (staggered rows of four at $r = 1/9$, sum 2.2222222, against the family best $V(4,4) = 2.2071068$; both valid at 10^{-9}, all 15 of that cell's valid rows are) and one at $N = 30$ ($r = 1/11$, sum 2.7272727, against $V(5,5) = 2.7071068$; valid at 10^{-6}, not at 10^{-9}, so it does not clear) — hexagonal layouts the family does not contain, found by a post-hoc sweep (not preregistered) run after a fourth-round check; they are not the modal output at either cell and do not bear on the discriminating-cell claims. Regenerates from `arm_m_collect.jsonl` via `arm_m_analysis.py`.

The net effect on the paper's claim is a sharpening. Arm M does not rescue $V(k,m)$; the closed form now reads "$k^* = \mathrm{round}(\sqrt{N})$ names the grid family; at $k^{*2} > N$ the model truncates that grid (five k confirmed); at $k^{*2} \leq N$ it extends only to $m \leq 1$ — beyond that it prefers the $(k^*{+}1)$ rectangle, and $V(k,m)$ does not describe it."

S5 Rectangle transfer (arm G)

S5.1 Transfer of the rule

For a $1 \times a$ rectangle the template has two parameters: $q^* = \mathrm{round}(\sqrt{N/a})$ columns across the width, $p^* = \mathrm{round}(\sqrt{N \cdot a})$ rows across the height. At $a = 1$ both collapse to $\mathrm{round}(\sqrt{N})$ — the same rule with the aspect ratio put back, not a new rule fitted to new data, and no rectangle model output existed when it was written. It was verified against an independent LP at 213 configurations, drift below 10^{-9}, including a

Table S1: Forecast versus outcome, both containers.

Cell	Branch	Predicted	Rival argmax	Valid / sampled	On-pred / valid	Rival
N=13 (sq)	truncate	1.6250000	1.7761424	4/5 · **18/20**	3/4 · **10/18**	0
N=17 (sq)	extend	2.0517767	†	4/5	3/4	†
N=21 (sq)	truncate	2.1000000	2.2588835	7/10 · **15/20**	4/7 · **12/15**	2
N=31 (sq)	truncate	2.5833333	2.7485281	5/5 · **17/20**	5/5 · **13/17**	0
N=35 (sq)	truncate	2.9166667	†	4/5	3/4	†
N=37 (sq)	extend	3.0345178	†	4/5	3/4	†
N=43 (sq)	truncate	3.0714286	3.2416246	7/10	6/7	0
N=19, a=3	truncate	3.1666667	3.5749194	4/8	2/4	0
N=25, a=2	truncate	3.1250000	3.4832492	7/8	3/7	0

Plain entries are the original 45-invocation bare arm (Appendix E.3's corpus); **bold** the same cells over the full 85-row bare ledger after §S6 added 40 rows. † marks non-discriminating cells (prediction equals family argmax), excluded from rival denominators. Square discriminating cells: 18/23 and 2/23 on the original corpus; 41/57 and 2/57 on the full ledger. Rectangle: 5/11 and 0/11. Sources: `arm_f_repro.py` over `arm_f_candidates_v2.jsonl`, `arm_g_rect.py` over `arm_g_candidates.jsonl`.

cross-domain check: at $a = 1$ with $p = q = k$ the square file's closed form must equal this file's LP. Shape mismatch grows as a leaves 1 — over $N = 10 \ldots 45$ the count of N whose predicted shape differs from the optimal one rises from 12 at $a = 1.0$ to 23 at $a = 1.5$, 20 at $a = 2.0$ and 23 at $a = 3.0$.

We probed the two sharpest cells with sixteen invocations (eight per cell) in a container none had been given before: $N = 19$ at $a = 3$ (predicted 3.1666667, rival 3.5749194) and $N = 25$ at $a = 2$ (predicted 3.1250000, rival 3.4832492). With all sixteen scored, **5 of 11 valid proposals landed on the predicted value and 0 of 11 reached the rival**.

We characterize all eleven here. On-prediction: two at $N = 19/a = 3$ (3.1666666667 and 3.16666673, the latter agreeing to six decimals not seven) and three at $N = 25/a = 2$ (3.125 exactly). Off-prediction at $N = 25/a = 2$: two samples emitted a uniform 5×5 grid at $r = 0.1$ summing to 2.5000000 — the *square* rule with no aspect correction, $k = \text{round}(\sqrt{25}) = 5$ — plus one at 3.0 and one at 3.151875, the latter above the 3.125 prediction from outside the family and below the rival. At $N = 19/a = 3$, **two** further samples exceeded the prediction from outside the family, 3.45 and 3.5, both below the rival.

We therefore report the rectangle as **partial out-of-sample support, not confirmation**: $5/11 = 45\%$, Wilson 95% CI $[21\%, 72\%]$, and a proposer choosing uniformly among the three or so plausible template shapes would land on-prediction about a third of the time, so 5/11 does not separate cleanly from that null. Two qualifications: validity degrades in the tall container (4/8 at $a = 3$ against 7/8 at $a = 2$, three of four $a = 3$ failures being overlaps), a separate finding and a confound on that cell at $n = 8$; and the rectangle ledger carries no prompt hashes, no run dates and no proposer fields, so the arm with the strongest transfer claim has the least provenance (§8).

S5.2 Negative result: the closed form does not survive the move to rectangles

The rectangle generalizes the *rule* but not the *formula*, and we keep the failure because it is what a verification gate is for. In the square every interior vertex has four identical neighbors, so one expression covers every filler. In a $1 \times a$ rectangle, with half-spacings $h_x = 1/(2q)$ and $h_y = a/(2p)$, a filler is also capped by horizontal and vertical spacing to *adjacent fillers*. Those constraints are inactive when $h_x = h_y$ — why the square case could not have revealed them — and bind only when neighboring vertices are occupied, so the cap depends on m and on which vertices a construction uses, and no expression in (p, q, m, a) reproduces it. The LP gate caught this on the first run: at $a = 1$, $p = 2$, $q = 4$, $m = 1$ the natural generalization $\texttt{rf} = \min(\texttt{diag}, h_x, h_y)$ returns **1.125** against a true **1.1545085**, capping against a neighboring filler that does not exist. We retain closed forms only where provably exact (full grid and truncated grid) and use the LP as the value oracle for the extend branch, so what §S5.1 tests is the LP-backed prediction.

S6 Elicitation and faithfulness (arm T)

This arm is secondary to the paper's question and its result leans negative: the registered validity prediction fails, the concentration effect survives only as a fragile directional finding, and the faithfulness audit checks description against artifact, not against process.

S6.1 Design and bundling disclosure

A ten-versus-ten pilot at $N = 21$ compared the bare prompt against a variant asking the proposer to name its construction on a leading `METHOD:` line: validity rose from 7/10 to 10/10 and the higher-scoring rival disappeared, from 2 of 7 valid bare samples to 0 of 10 trace samples. Before any scaled sample was drawn we registered four predictions and an explicit falsifier in `arm_t_preregistration.txt` (sha256 `ab7900a8...`), disclosing that the pilot's trace prompt had drifted from the bare template beyond the method line — it omitted the `[0,1]x[0,1]` tokens and reworded the output-format line — so the pilot's intervention was method-line-plus-rewording, bundled. Pilot samples are never pooled with `trace_v2`.

The scaled prompt, `trace_v2`, is itself a **bundled prompt-format-and-trace-request** intervention, though a near-minimal diff: one inserted `METHOD:` line, plus `"After the METHOD line, "` prepended to the output line and `"no other text"` changed to `"no other text after the list"`. That second change is not cosmetic — §S6.2 shows it doing measurable work — so the measured effect is *method-line-plus-output-line-rewording*, not the method line alone: a weaker version of the confound for which the pilot was discarded, held to the same standard.

The scaled arm ran 100 new invocations, bringing both arms to 20 per cell at $N \in \{13, 21, 31\}$. The study corpus accumulates as tabulated below. Each arm is scored under its own registration; the running total is the figure quoted elsewhere in the paper, and with arms P, CL, CCP, CL-W and P-D the study corpus now stands at 1500 invocations.

arm or group	section	invoca
square (bare 85, trace 70, sonnet 30, `opus_alias` 30)	§3.3, §S6	
rectangle	§S5	
M	Appendix E.4	57 sampled of 60 laun
MU and CH	§5.1, §4.1	180, non
CC, CC2 and CCS	§3.4	135, non
CN, held-out N	Appendix E.5	
CP, RP and PP	Appendix E.6	270, three rejections cou
L, iterated loop	§5.2	120, two rejections cou
P, pinned path	§5.3	225, six refused by the serving
CL, library code	§3.5	90, four refused by the serving
CCP, isolating arm	§3.5	45, one in-flight refusal re
CL-W, weak-tier top-up	§3.5	45, non
P-D, runtime component	§5.3	27 of 30 collected, closed there by amendment, three refused by c

The square row is 85 bare (45 original + 40 new), 70 trace (10 pilot + 60 trace_v2), 30 sonnet and 30 `opus_alias`. The CP/RP/PP row is 75 offset-container, 90 recall, a 15-invocation registered positive control and 90 paraphrase invocations. The cross-vendor chain sits outside this ladder because it is scored under its own registrations: 420 GM-chain invocations (§5.4, §7) and 877 arm-V attempt rows over the 325 registered invocation slots, 314 of which carry rows (§5.4). One disclosure: **20 of the 60 bare samples are pre-existing arm-F rows** ($N = 13$ ids 1–5, $N = 21$ ids 1–10, $N = 31$ ids 1–5) whose results were known when P-T1–P-T3 were written; only trace_v2 is fully fresh (§S6.4). Scoring used `arm_f_repro.py` unchanged with the registered 2×10^{-3} window; the Fisher exact test comes from the hypergeometric tail in `arm_t_analysis.py`, checked against a reference.

Table S2: Paired trace grid (scaled arm only).

N	Arm	n	Valid (parse / geom. fail)	On-pred / valid	Rival
13	bare	20	18 (0 / 2)	10 (56%)	0
13	trace_v2	20	18 (0 / 2)	16 (89%)	0
21	bare	20	15 (1 / 4)	12 (80%)	2
21	trace_v2	20	18 (0 / 2)	16 (89%)	0
31	bare	20	17 (2 / 1)	13 (76%)	0
31	trace_v2	20	17 (0 / 3)	14 (82%)	1
pooled	bare	60	50 (3 / 7)	35 (70%)	2
pooled	trace_v2	60	53 (0 / 7)	46 (87%)	1

S6.2 Registered outcomes

Preregistered criteria are evaluated exactly as registered, and unregistered alpha thresholds are applied nowhere.

P-T1, validity: NOT confirmed. P-T1 is the one prediction that registered an alpha ("validity $\geq$ bare at EACH N, and pooled one-sided Fisher $p < 0.05$"). Direction held at all three cells, but the pooled test gives $p = 0.30$: 53/60 (88%) trace_v2 against 50/60 (83%) bare, a difference whose detection needs several hundred samples per arm at 80% power — a failure to detect, not a demonstration of no effect. The direction that did hold is not geometric: bare had **3 parse failures and 7 geometric failures**, trace_v2 **0 parse and 7 geometric**, so among parsed samples geometric validity is 50/57 (87.7%) versus 53/60 (88.3%). The P-T1 direction is format compliance, exactly what the reworded output-format line manipulates; validity is reported split by cause from here on.

P-T2, rival suppression: CONFIRMED as registered (directional, no alpha): 1 of 53 valid trace_v2 against 2 of 50 valid bare. Counts this near zero carry no inferential weight and we claim none (Fisher $p = 0.48$, carrying no decision). One trace_v2 sample at $N = 31$ hit the rival 2.7485281 to within 3.6×10^{-8} — the first weak-tier rival hit at $N = 31$ in any arm, and formerly one of the scorer's mismatches (§S6.5).

P-T3, anchor concentration: met as registered (directional), inferentially fragile. Among valid samples, 46 of 53 trace_v2 (87%) landed on the registered prediction against 35 of 50 (70%) bare. The p-value it never carried is 0.0325, one-sided uncorrected (§S6.4).

P-T4, faithfulness: confirmed under a corrected scorer. The registered scorer gives 38 of 41 scoreable claims matching (93%) against a 90% threshold; blind hand-adjudication under a rubric frozen before rescoring gives 54 of 56 (96.4%). §S6.5 reports both.

Per-cell one-sided Fisher on on-prediction (pooled bars in Figure S2): $N = 13$ $p = 0.030$; $N = 21$ $p = 0.41$; $N = 31$ $p = 0.50$. Pooled $p = 0.0325$; Holm threshold over the registered family ($m = 3$ tested with p-values) is 0.0167. The pilot (10 invocations at $N = 21$, 10/10 valid, 9/10 on-prediction) is reported separately and never summed into any total. Regenerated by `arm_t_analysis.py` over `arm_f_candidates_v2.jsonl`.

S6.3 The registered falsifier, both readings

The registered falsifier reads: *"if trace_v2 validity* `<=` *bare at 2+ of 3 N, the pilot effect was the bundled rewording, not the method-line request — reported as such, not reframed."* Observed validity: $N = 13$, 18/20 both arms — **tie**; $N = 21$, trace_v2 18/20 against bare 15/20 — trace higher; $N = 31$, 17/20 both arms — **tie**. A tie satisfies `<=`, so under the registered wording the falsifier is **triggered at 2 of 3** N, and we report it as triggered; the code instead evaluated direction as `t_rate >= b_rate`, strict `<`, under which it fires at 0 of 3 cells. That convention appears nowhere in the preregistration, was chosen at analysis time, and read the same tie favorably twice — as "direction held" for P-T1 and as "not a falsifier cell" here.

We name the failure mode **operationalization drift between preregistration text and analysis code**: registered prose leaves boundary behavior — ties, inclusive versus exclusive bounds, rounding — implicit,

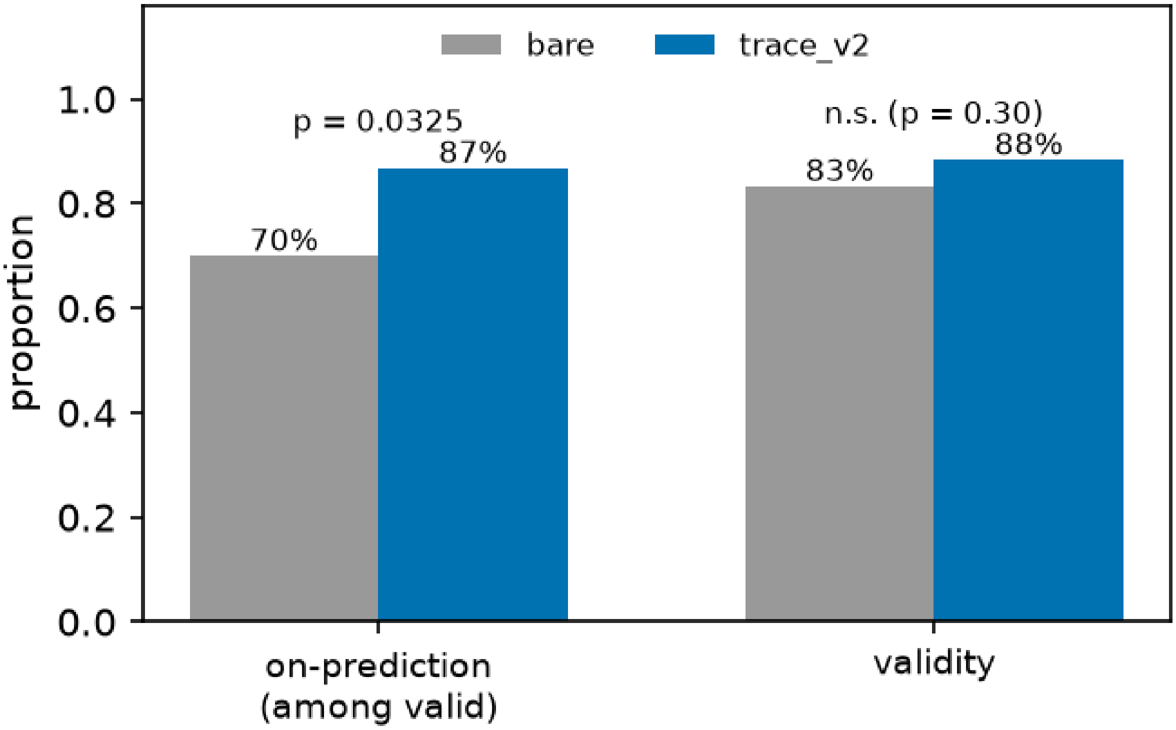

Figure S2: *Bare versus* `trace_v2`*, pooled over cells.* The annotated $p = 0.0325$ is uncorrected and fails the Holm threshold (§S6.4); per-cell values are in the text. Caption caveat: the bars pool the bare arm's two collection waves, which §S6.4 shows drift materially, so the pooled bare bar is a mixture; the same-wave comparison in §S6.4 is the wave-controlled picture.

and the scorer must choose after the data exist; the preregistration literature (§6) discusses what is locked, not this tie-handling layer. Remedies, free before sampling and impossible afterward: register the comparison as executable code, or state the tie convention in the registration text. `arm_t_analysis.py` now prints both readings and names the registered tie-inclusive one authoritative. Under that reading **the pilot's validity effect is attributed to the bundled rewording, not to the method-line request**, as the falsifier clause instructed; the §S6.2 parse-versus-geometric split points the same way, and P-T1 fails under both readings regardless.

S6.4 Limits of P-T3: what the concentration result will and will not carry

P-T3 met its registered directional criterion. It is not a flagship result.

No registered alpha. P-T3 registered direction only; the p-value is a post-hoc addition.

Multiplicity. Of four registered predictions, three are reported with p-values. Holm sets the first threshold at $0.05/3 = 0.0167$; $p = 0.0325$ fails it, as it fails Bonferroni at any $m \geq 2$ and Benjamini–Hochberg FDR at $q = 0.05$. Two-sided Fisher is $p = 0.0537$. P-T3 is nominally significant uncorrected and is not under family-wise error control.

Concentration in one cell. Per-cell tests give $p = 0.030$, 0.41, 0.50 (Table S2); the pooled 17-point gap is a 33-point gap at $N = 13$ averaged with 9- and 6-point gaps elsewhere, driven by a *low bare rate at* $N = 13$ ($10/18 = 56\%$) rather than a high trace rate — the trace rate there (89%) equals that at $N = 21$. Dropping $N = 13$, the comparison is 30/35 vs 25/32, $p = 0.31$.

Wave confound. The bare arm mixes two collection waves; trace_v2 is one. Splitting bare on the registered id boundary, with no intervention applied to either half, on-prediction is: $N = 13$, 3/4 old wave vs 7/14 new; $N = 21$, 4/7 vs 8/8; $N = 31$, 5/5 vs 8/12 — the control arm's own between-wave drift is comparable to the effect attributed to the manipulation. (An earlier draft printed 2/4 vs 10/11 at $N = 21$ and a 25/37 pooled figure — arithmetically impossible against §S6.1's id boundary; corrected here from the ledger.) Restricting to one wave, new-wave bare is 23/34 (68%) against trace_v2's 46/53 (87%), but **this stratified comparison is post hoc, not preregistered, no confirmatory weight** — a diagnosis, not a rescue: wave is a confound *in the registered analysis*, which cannot separate it from the manipulation. The design fix, interleaved collection in one session, was not run.

What we therefore claim. A minimal method-naming request does not make the proposer better at geometry and does not measurably change which rare constructions it reaches; in the registered direction it concentrates output onto the anchor, but that concentration fails correction, is carried by one cell, is confounded with collection wave, and is bundled with an output-format rewording that demonstrably affects parse compliance. Studies collecting process descriptors *by asking for them*, including descriptor-driven quality-diversity pipelines where the descriptor is a requested self-report, should treat this as a reason to check, not a coefficient to apply. The pilot also showed a *third* effect in the P-T3 direction at the same cell (trace 9/10 on-prediction against bare $N = 21$ at 4/7, one-sided $p = 0.16$), so P-T3 replicated a pilot signal rather than testing a blind prediction — a different status, and the prereg's pilot disclosure covered the prompt drift only.

S6.5 Faithfulness with ground truth

Claim and artifact sit in the same completion and the artifact is fully determined, so a method line is checkable against the emitted coordinates. The original automated scorer, `arm_f_repro.trace_faithfulness()`, returned 38 of 41 scoreable claims matching (93%) against the registered 90% threshold, but nine of its twelve exclusions were regex coverage gaps rather than claims without numeric content, and the excluded set is enriched for non-canonical phrasings where a mismatch is a priori *more* likely — so the exclusion was not conservative in a known direction. Worst case, had all twelve been unfaithful, the rate would have been $38/53 = 72\%$. §S6.6 lists the twelve exclusions verbatim and names the two regexes that produced them. That accounting is why the arm was rescored blind.

Blind rescore under a frozen rubric. The rubric (`p7_faithfulness_rubric.md`) — checkable assertions in any surface form, plus match rules including a base-grid convention for "grid plus additions" claims and a truncation convention for "with k removed" claims — was committed to git as `e181d2a` at 03:56:56 on 2026-08-01, **before** any rescoring run, so the freeze is checkable, and it pre-committed the reporting rule: below 90%, P-T4 reported not confirmed, full stop, no consolation framing. Scoring was blind — claim text and circles only, no predicted or rival values, no running tally, no threshold and no sight of the existing 38/41 result — with tallying afterward; all 60 labels are released verbatim in `p7_blind_labels.json`.

Result. Over all 60 trace_v2 rows: **54 MATCH, 2 MISMATCH, 4 UNSCOREABLE** — a corrected rate of $\mathbf{54/56 = 96.4\%}$ of scoreable claims, Wilson 95% CI $[88\%, 99\%]$, passed, and on 56 scoreable claims rather than 41. The CI's lower bound sits just below the registered threshold, so "clears 90%" is a statement about the point estimate at this n. The two mismatches share a form: row 11 claims "alternating rows of 4 and 3 circles" over observed row sizes 4,3,3,3, row 33 claims "5 rows alternating between 5 and 4 circles" over observed 5,4,4,4,4 — both *mis-descriptions of alternation*, a repeating pattern named and built for the first period only. The original scorer's three mismatches were three different claim forms, not one; the filler-adds-rows mechanism the original scorer described was real, and the rubric's base-grid convention handles it, so all three score MATCH under hand adjudication.

Scope, three ways. The blind pass covers *invalid* rows too: the original 93% conditioned on completions that produced a correct packing — the wrong conditioning, since a method line attached to an overlapping layout is where description and artifact are likeliest to diverge — while the blind pass scored all 60 rows including the 7 invalid ones, 4 of the 60 unscoreable. Second, the check verifies the description against the *artifact*, not the process: the non-claim guard applies in full, and for a model already emitting grids, "5×5 grid" matching a 5×5 grid is closer to a self-consistency check than to the causal question the chain-of-thought literature asks (§6). Third, the audit is confounded with the elicitation arm, computed on the arm §S6.2 shows concentrates 87% of valid outputs onto one template, where correct description is close to free; faithfulness is informative precisely on off-template outputs, such as the pilot's triangular-hexagonal 6+5+4+3+2+1 sample, described exactly while summing to 1.75. The split by on- versus off-prediction is not run here (§8, stopping rule). What survives is narrow: in a domain where claims are checkable, the method lines this proposer emits are, at 54 of 56 scoreable claims, true of the object it produced — and both failures are a model describing a pattern it started and did not finish.

S6.6 The twelve exclusions of the original faithfulness scorer

§S6.5 states the consequence. The automated scorer's twelve exclusions were not claims lacking numeric content. Nine were regex coverage gaps: `DIMS_RE` matched only the literal `NxN` form, and `ROWSCOLS_RE` required rows before columns, so the following fell through despite carrying explicit checkable dimensions.

- "Regular 3 by 4 grid with one additional circle in row 4"
- "3 complete rows of 4 circles and 1 centered circle in the top row"
- "rectangular grid 4-4-4-1 with uniform radius 1/8"
- "Four-row grid with 4 circles in the first row and 3 in each of the remaining"
- "Rectangular grid arrangement with 5 columns and 4 rows, plus one additional circle on top edge"
- "Square grid 5 columns 4 rows with 1 additional circle at top"
- "Rectangular grid six columns by six rows radius one-twelfth with five circles removed"
- "Regular 5-column by 6-row grid plus one additional circle in the right margin"
- "Rectangular grid packing with five columns and six rows plus one additional circle"

Three further exclusions are closer to genuinely unscoreable: "Uniform horizontal strips with tight row spacing"; "Regular rectangular grid packing with corner circle"; "Equal-radius rectangular grid with 4 rows stacked vertically".

The nine gap cases are non-canonical phrasings, which is where a description is a priori *more* likely to diverge from the artifact. An exclusion rule that drops exactly those is not conservative in a known direction, and that is the reason §S6.5 reports the blind rescore over all 60 rows rather than the original 38 of 41.

S7 Arm B: the reference program, verbatim

The optimizer-alone baseline of §3.5 is one fixed program. It is reproduced here so that the claim that it is what a reader would write first can be checked rather than taken on assertion. Authorship is the manuscript's (Claude drafting under the human author, §8); version 1 was committed at `1932cc0` and this version, version 2, at `c486843`, both before the single scored run, and no result informed either. The runner substitutes `__N__` and `__SEED__` and hands the source to arm CL's scorer as if it were a model completion; the report asserts the template's SHA-256, `298ba71c9f20...`. Version 1 differed in two places only: it bounded restarts by wall clock, importing `time` and `sys`, which the registered allowlist forbids, and was blocked at all 45 rows; version 2 fixes the count at 50.

```
# Arm B: the optimizer alone. A fixed reference program, written by the pipeline before any
# run and never tuned on a result, that does what a reader would write first: random-restart
# SLSQP over centre and radius variables, maximising the sum of radii under containment and
# pairwise non-overlap, for a fixed number of restarts. It is executed through arm CL's
# registered pipeline unmodified (python -I -S, the fixed driver, 120-second wall clock, one
# core, arm-F scoring, section 2.4's clearance rule), so its rows are scored exactly like the
# model-written programs it is the baseline for. Imports are the registered allowlist only
# (math, numpy, scipy); version 1 imported time and sys and was blocked (amendment 1).
#
# Usage inside the pipeline: arm_b_run.py substitutes N and SEED below and hands the source
# to arm_cl_analysis.score_row as if it were a model completion.
import numpy as np
from scipy.optimize import minimize
```

```python
N = __N__
SEED = __SEED__
RESTARTS = 50   # amendment 1: fixed count; 80 restarts fit in 95 s at N = 31 on one core

rng = np.random.default_rng(SEED)
iu = np.triu_indices(N, 1)


def unpack(v):
    return v[:N], v[N:2 * N], v[2 * N:]


def objective(v):
    return -np.sum(v[2 * N:])


def objective_grad(v):
    g = np.zeros_like(v)
    g[2 * N:] = -1.0
    return g


def constraints(v):
    x, y, r = unpack(v)
    dx = x[iu[0]] - x[iu[1]]
    dy = y[iu[0]] - y[iu[1]]
    rs = r[iu[0]] + r[iu[1]]
    return np.concatenate([x - r, 1 - x - r, y - r, 1 - y - r, dx * dx + dy * dy - rs * rs])


def constraints_jac(v):
    x, y, r = unpack(v)
    m = len(iu[0])
    J = np.zeros((4 * N + m, 3 * N))
    ar = np.arange(N)
    J[ar, ar] = 1.0;            J[ar, 2 * N + ar] = -1.0
    J[N + ar, ar] = -1.0;       J[N + ar, 2 * N + ar] = -1.0
    J[2 * N + ar, N + ar] = 1.0;  J[2 * N + ar, 2 * N + ar] = -1.0
    J[3 * N + ar, N + ar] = -1.0; J[3 * N + ar, 2 * N + ar] = -1.0
    i, j = iu
    rows = 4 * N + np.arange(m)
    dx = x[i] - x[j]
    dy = y[i] - y[j]
    rs = r[i] + r[j]
    J[rows, i] = 2 * dx;        J[rows, j] = -2 * dx
    J[rows, N + i] = 2 * dy;    J[rows, N + j] = -2 * dy
    J[rows, 2 * N + i] = -2 * rs; J[rows, 2 * N + j] = -2 * rs
    return J


bounds = [(0.0, 1.0)] * (2 * N) + [(0.0, 0.5)] * N
cons = [{"type": "ineq", "fun": constraints, "jac": constraints_jac}]


def repaired(v):
    """Shrink every radius by the largest constraint violation so the packing is strictly
    feasible; SLSQP satisfies constraints only to its own tolerance."""
    x, y, r = unpack(v.copy())
    viol = max(0.0, -np.min(constraints(v)))
    r = np.maximum(r - viol - 1e-12, 0.0)
    return x, y, r


best_sum, best = -1.0, None
for _ in range(RESTARTS):
    v0 = np.concatenate([rng.uniform(0.1, 0.9, N), rng.uniform(0.1, 0.9, N), rng.uniform(0.02, 0.08, N)])
    res = minimize(objective, v0, jac=objective_grad, method="SLSQP", bounds=bounds,
                   constraints=cons, options={"maxiter": 400, "ftol": 1e-12})
    x, y, r = repaired(res.x)
    if np.min(constraints(np.concatenate([x, y, r]))) < 0 or np.any(r <= 0):
        continue
    s = float(np.sum(r))
    if s > best_sum:
        best_sum, best = s, (x, y, r)

if best is None:
    print("[]")
else:
    x, y, r = best
    print("[" + ", ".join(f"[{a:.10f}, {b:.10f}, {c:.10f}]" for a, b, c in zip(x, y, r)) + "]")
```